

Requirement Analysis Phase
Solution Requirements (Functional & Non-functional)

Date	28 June 2026
Team ID	SWTID-2026-2260
Project Name	Smart Lender
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Loan Application	Enter applicant details through the web form. Validate all required input fields.
FR-2	Data Preprocessing	Handle missing values. Encode categorical data. Prepare input features for prediction.
FR-3	Loan Approval Prediction	Load the trained XGBoost model. Predict whether the loan is Approved or Rejected. Display prediction result instantly.
FR-4	Cloud Deployment & Result Display	Deploy the Flask application on IBM Cloud. Display prediction results through the web interface. Store prediction history (optional).

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
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NFR-1	Usability	The application should provide a simple, user-friendly interface that allows users to submit loan details and receive prediction results easily.
NFR-2	Security	User data should be validated and securely processed. The application should use secure communication and protect sensitive applicant information.
NFR-3	Reliability	The system should consistently generate accurate loan approval predictions with minimal errors and high availability.
NFR-4	Performance	The application should provide real-time loan predictions with low response time and efficient resource utilization.
NFR-5	Availability	The system should be accessible online at all times through IBM Cloud deployment with minimal downtime.
NFR-6	Scalability	The application should support future enhancements, additional users, and integration of new machine learning models without major changes.